

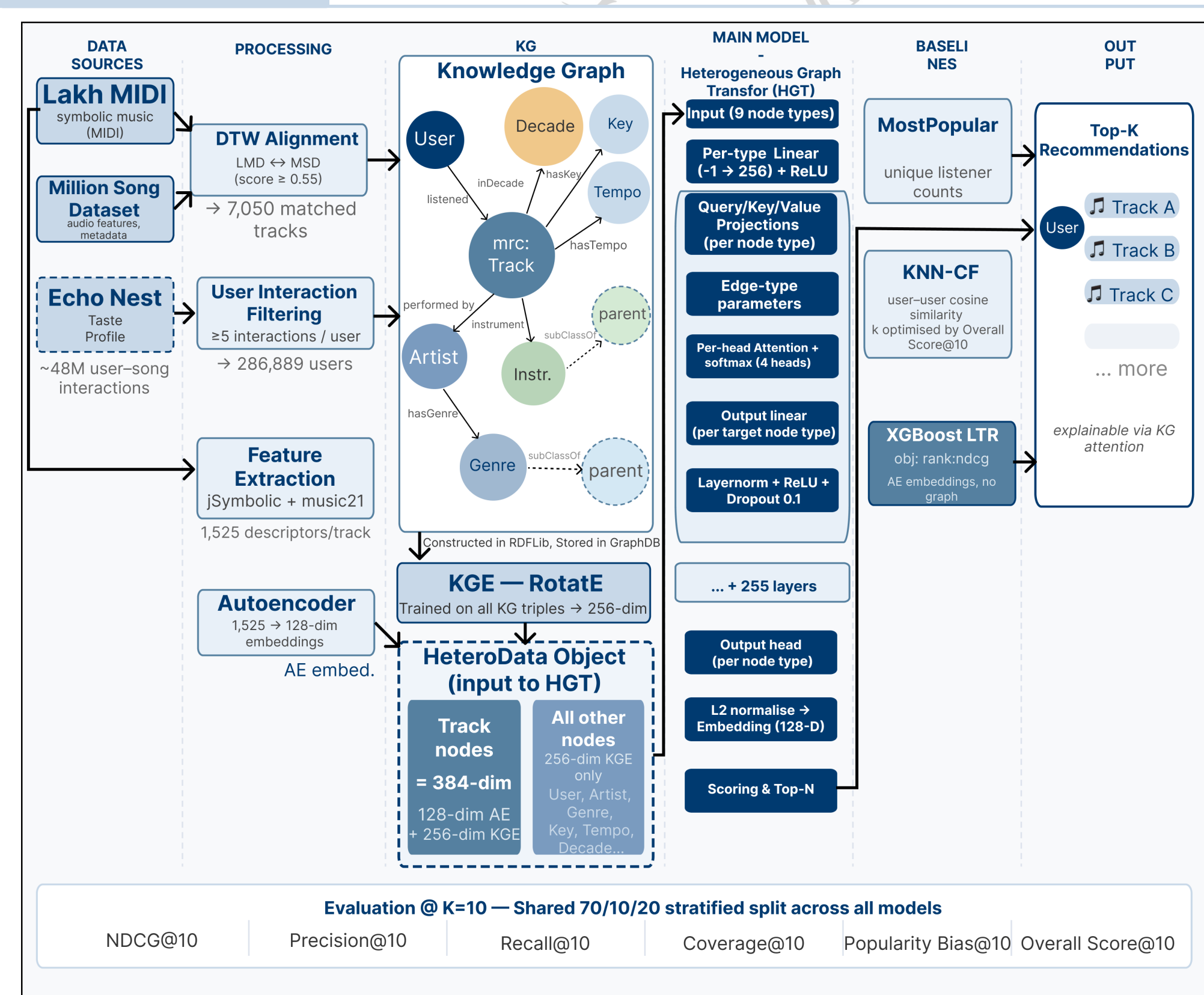
Objectives

- Recommend new music by enriching collaborative filtering (CF) with **symbolic + audio content** — genre, artist links, instrumentation, tempo.
- Encode it as a typed **Knowledge Graph** relating users, tracks, artists and genres beyond a user-item matrix.
- Learn over the KG with a **Heterogeneous Graph Transformer (HGT)** for diversity, cold-start robustness and **explainability**.

Problem

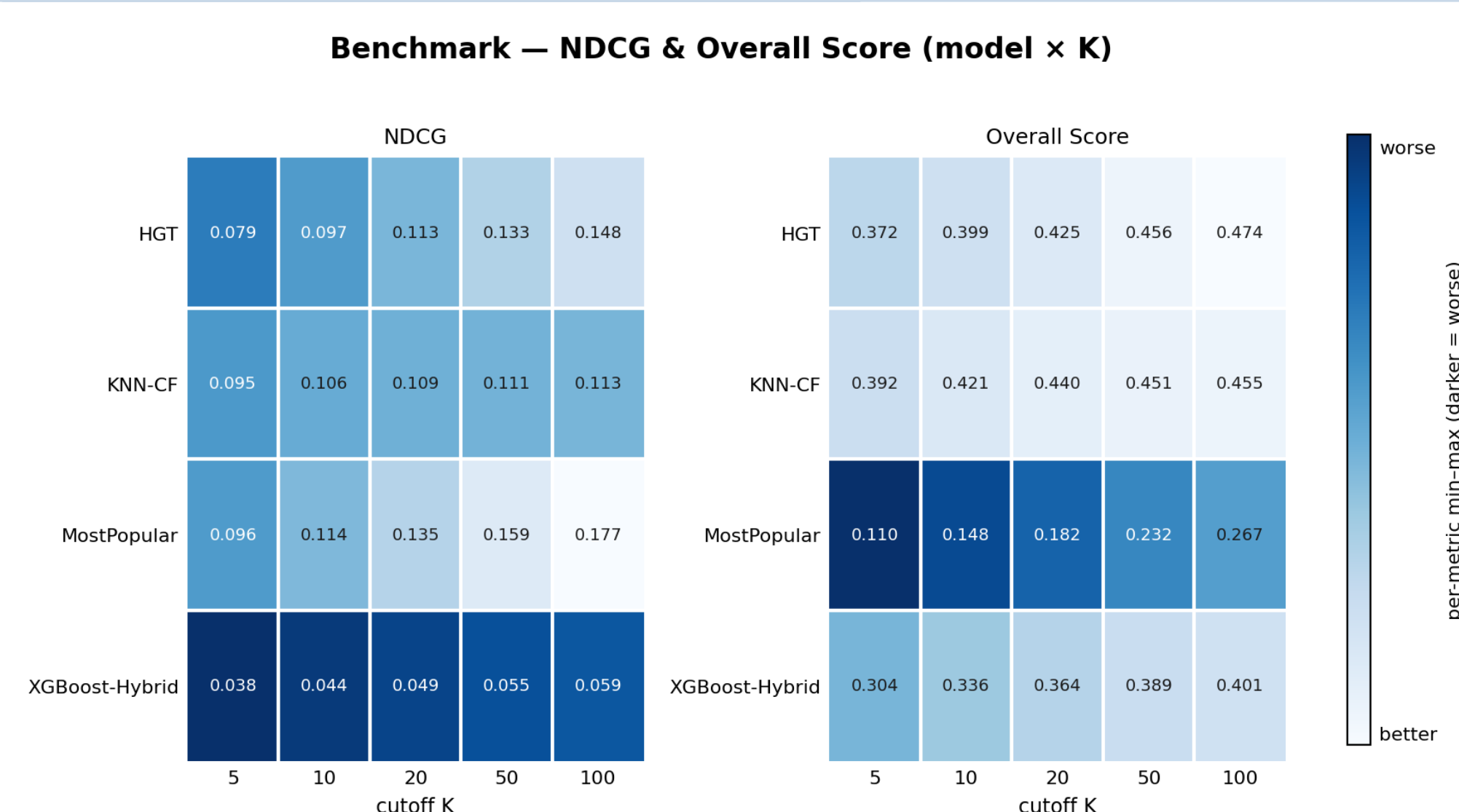
- CF uses only interaction history → **popularity bias** and **cold-start** failures, ignoring musical content.
- Most KG-enhanced work uses shallow embeddings [8, 10], without a GNN that propagates **multi-hop typed** structure.
- Can symbolic features + a heterogeneous KG + an HGT reason over shared genre / artist / instrumentation to close that gap?*

Methodology



Model selection accounts for relevance, diversity & fairness by using:
Overall Score@10 = $0.6NDCG + 0.2Cov + 0.2(1 - PopularityBias)$

Recommendation Benchmark

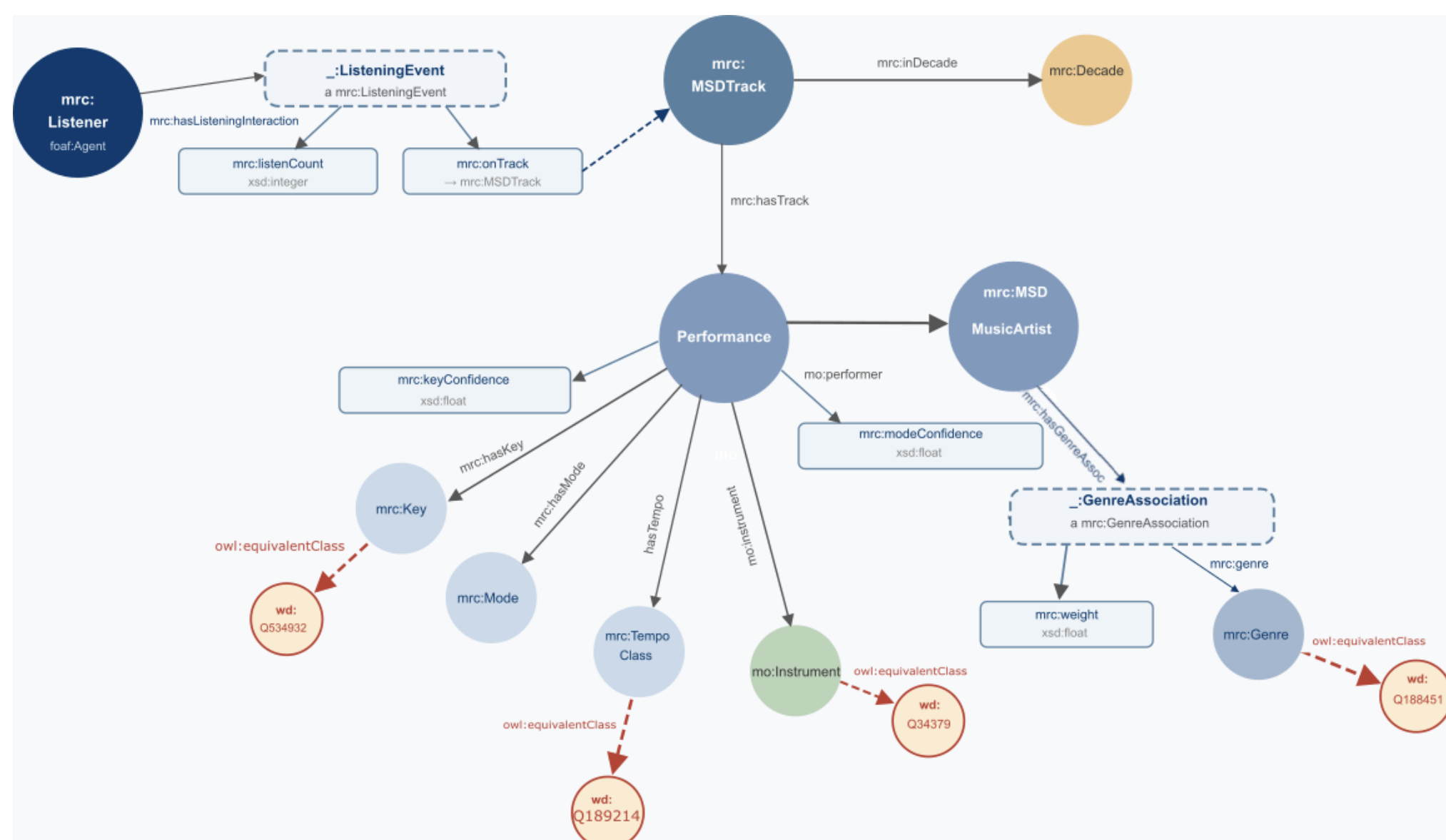


- HGT trails KNN-CF for short lists but **overtakes it for $K \geq 50$** : Overall Score **0.474 vs 0.455** and NDCG **0.148 vs 0.112** at $K=100$, with the **lowest popularity bias (0.090)** throughout.

Baselines

- MostPopular** — global play counts, no user features; rank by frequency → one list shared by all users.
- KNN-CF** — binarised, ℓ_2 -normalised interactions; user-user cosine k NN ($k=6$) → neighbour-weighted scores.
- XGBoost-Hybrid** — AE user+track embeddings + 44 categorical (1-hot key/mode/tempo; target-enc. genre/artist; multi-hot instrument); 200 AE-cosine candidates/user, gradient-boosted LTR re-rank (**rank:ndcg@10**) → top- K .

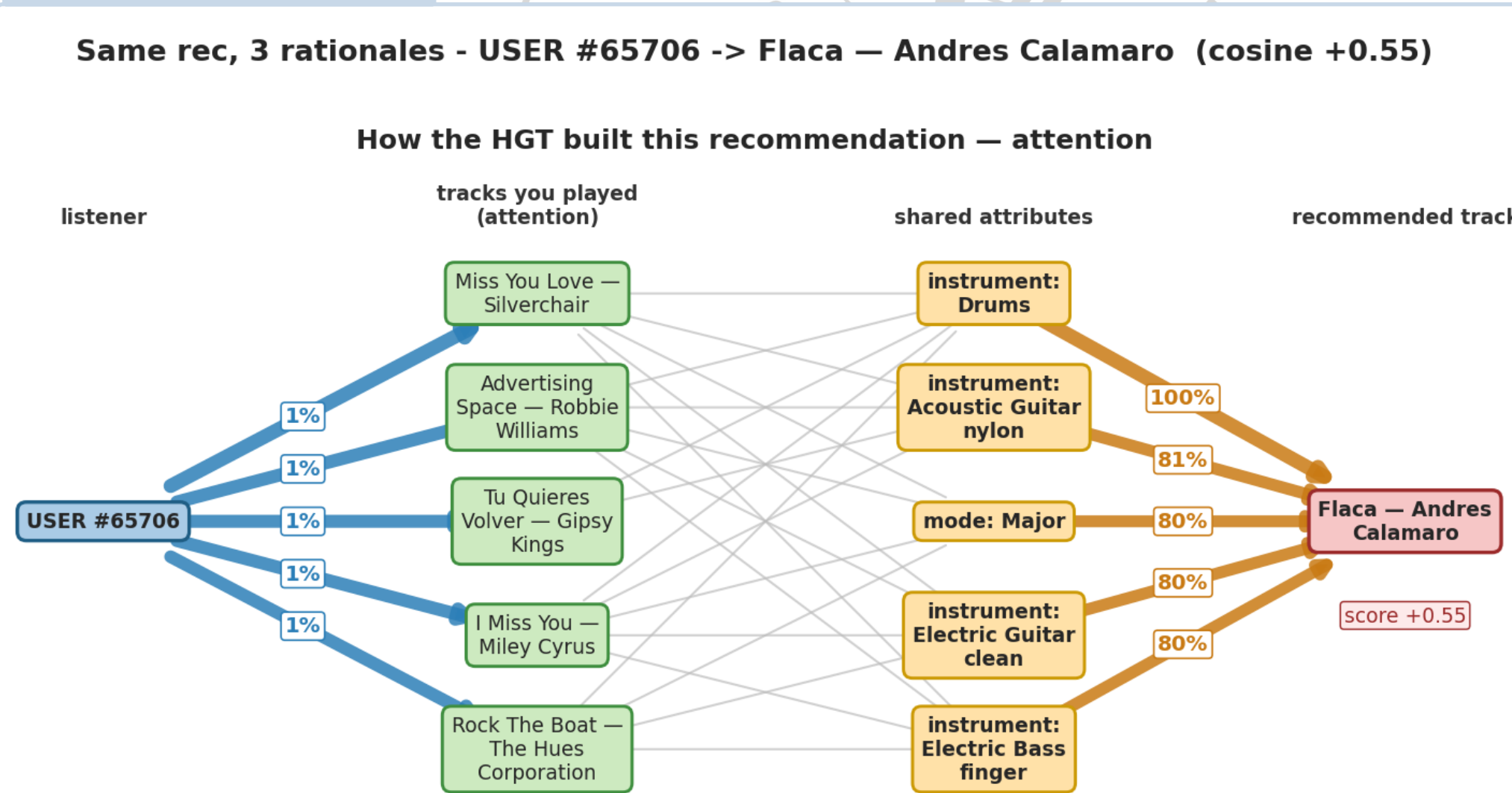
Knowledge Graph



TBox: ontology classes, properties and Wikidata alignments.

- 286,889** users, **7,050** songs (Lakh MIDI ↔ MSD, DTW ≥ 0.55); songs carry 1,525 symbolic descriptors + audio metadata.
- Genres / tempo / mode linked to **Wikidata** (**owl:equivalentClass**, **sameAs**); reified blank nodes encode weighted listening & genre events.

Explainability



Left edge: Each track's attention share (top 5 of 134). Right edge: Share of attention sitting on tracks carrying the attribute. Faithful attention traces users → tracks → shared attributes → recs.

- Captured **HGT attention** (not a post-hoc proxy) explains *why* each track was recommended, attribute by attribute.
- Cross-checked against **occlusion Δ score** and **Integrated Gradients** — the three agree on the top drivers.

Conclusions & Future Work

- The HGT **matches and overtakes** the strongest CF baseline at depth ($K \geq 50$) on NDCG, coverage & Overall Score, with the **lowest popularity bias** of all recommenders.
- Symbolic + audio content in a **typed KG** gives interpretable, diverse recommendations; brand-new users are served through **listener archetypes** (cold-start).
- Captured **HGT attention** yields faithful, attribute-level explanations, corroborated by occlusion Δ score and Integrated Gradients.

Future work

- Adaptive popularity control**: learn / anneal the debiasing weight λ and temperature τ to trade relevance against fairness on demand.
- A **hybrid HGT + CF** re-ranker that exploits the depth crossover — CF for short lists, the HGT for long, diverse ones.
- Richer, multi-hop KG signals** (lyrics, mood, social ties) and deeper propagation beyond the single user → track hop.
- Temporal / session-aware** modelling, plus a human study of how useful the attention explanations are to listeners.